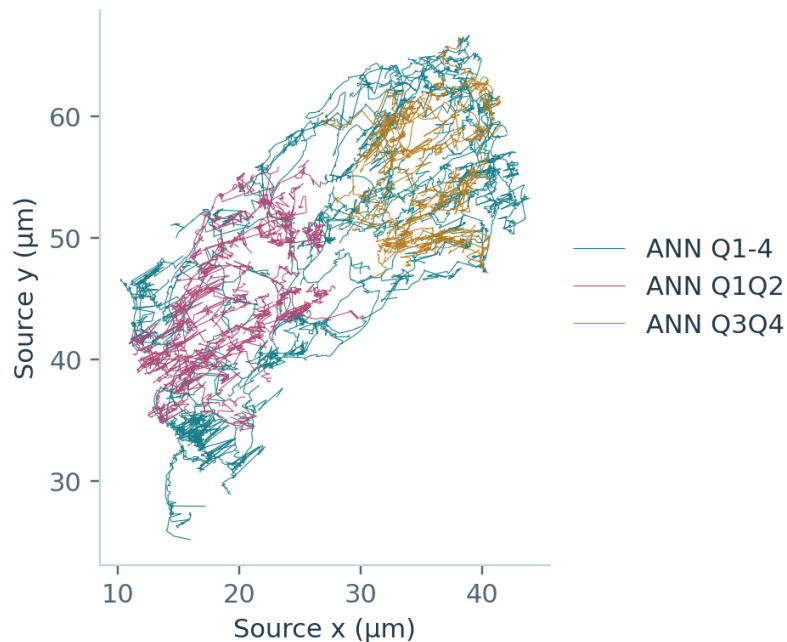


CombCircuit Live

The complete guide

Controls, parameters, signals and experiments



Measured ANN skeletons, projected onto the source x-y plane.

Explore any selected cell or ANN location, inject current, change the electrical hypothesis and compare spatial with equipotential responses.

Measured anatomy. Assumed physiology.

This guide distinguishes what the data establish from what the model assumes.

Start here

This guide explains **CombCircuit Live 0.2**, including the underlying model and the earlier fixed comparison included in the same download. Read pages 3-8 first, then use the parameter pages while you experiment.

Topic	Pages
The organism, circuit and two model representations	3-4
Arbors, live controls, probes, inputs and graphs	5-8
Every adjustable electrical parameter	9-11
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Live versus legacy behavior; Python parameter names	20
Troubleshooting, glossary and units	21-22
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Open the simulator

Unzip **CombCircuit-Live-v0.2.zip** and double-click **CombCircuit/LIVE.html**. A JavaScript-enabled browser with Web Workers is required. No Python installation, account, server or internet is needed to explore. Chromium was tested for this release; Safari and Firefox were not tested here.

Your first minute

Press **Run**. Keep the default demo pulse enabled, watch the clock pass 20-40 ms, then pause. Click an ANN branch to add a local probe. Change **Click action** to **Move injection site**, click another branch, and press **Inject now**. Use Run or Step to advance through that pulse.

Save before changing your measurement set. Export traces and the experiment JSON before resetting, clearing history, or adding/removing probes. The live app retains a rolling window, not an unlimited recording.

What is being modeled?

The organism is the comb jelly *Mnemiopsis leidyi*. The selected circuit comes from its **aboral organ**, associated with balance and coordination of ciliary activity. Aboral means away from the mouth. The original reconstruction was produced by Jokura and colleagues [1,2].

Component	Count	Meaning
ANN neurons	3	Aboral nerve-net neurons with explicitly modeled branched anatomy.
Bridge cells	26	Connected cells that include synaptic input returning to ANN neurons.
Balancer cells	129	Ciliated cells receiving modeled electrical input. Their ciliary mechanics are not simulated.
Selected cells	158	Includes balancer cells; this is not a count of 158 neurons.
Directed contacts	280	Measured sender-to-receiver contacts over 113 distinct directed cell pairs.

Quadrants and syncytial morphology

ANN Q1-4 spans four anatomical quadrants. **ANN Q1Q2** and **ANN Q3Q4** span two quadrants each. The names describe anatomical extent, not a proven hierarchy of control. ANN neurons have syncytial morphology, including branched continuous structures with multiple nuclei.

What is evidence, and what is assumed?

The model uses reconstructed paths, contact identities and contact directions. Cable radius, membrane properties, contact strength and sign are assumptions. The source ciliary measurements are bundled as potential future targets; they are not neural-voltage recordings and were not used to fit this model.

Scope: this is an induced ANN/bridge/balancer subgraph. Cells and influences outside that selection are omitted. Neither version is a whole-animal simulation.

Two representations of the same circuit

The research question is whether one voltage per cell is sufficient, or whether activity can vary meaningfully along a neuron’s reconstructed arbor.

	Spatial model	Equipotential / point model
ANN representation	Many connected electrical compartments per arbor.	One electrical compartment per ANN cell.
Total compartments	16,601: 16,446 ANN pieces plus 155 other cells.	158: one for every selected cell.
Internal voltage	Can differ from branch to branch.	Identical at all locations within a cell.
Local cable cut	Can disconnect a selected internal edge.	Cannot represent an internal local cut.
Matched quantities	Same total capacitance per cell, leak rule, input current and measured contacts.	The same matching applies; contact weights are not separately normalized.

A compartment is not another neuron

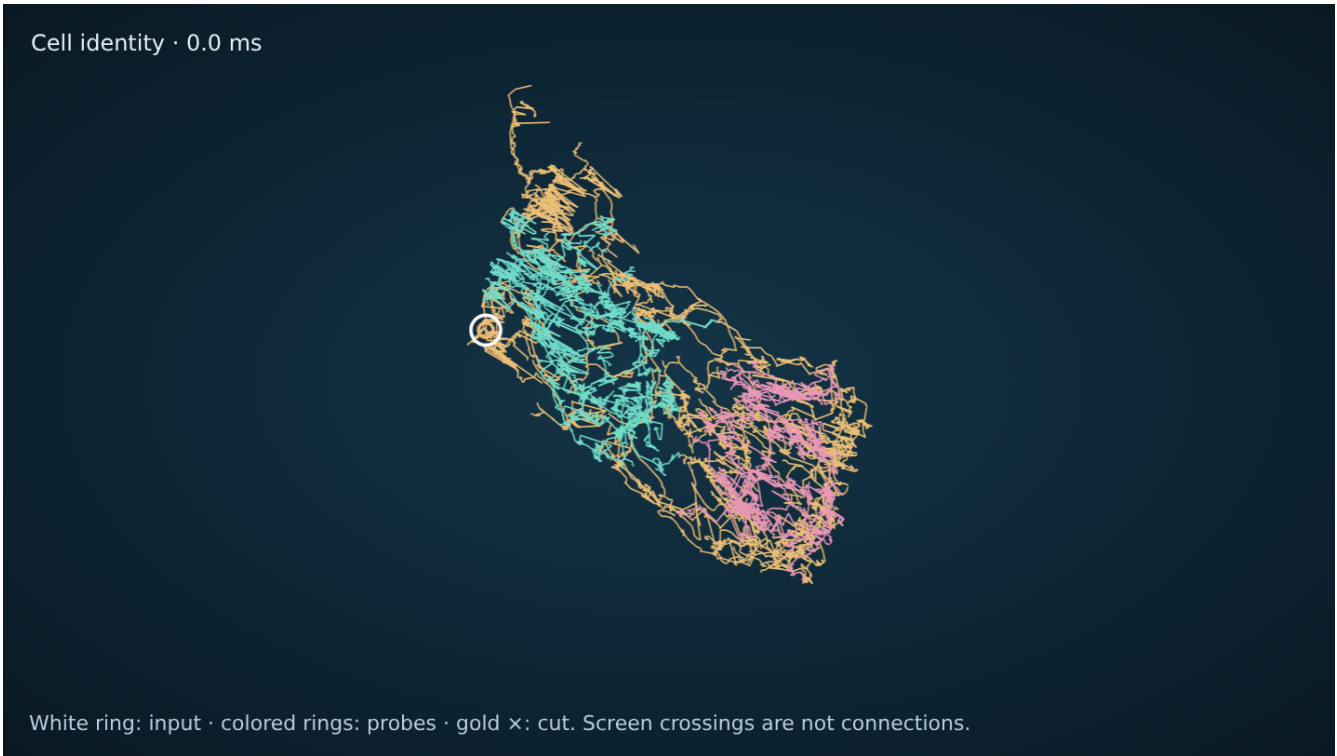
A compartment is a computational piece of a cell. Each stores charge, leaks current toward rest and exchanges internal current with adjacent pieces. Bridge and balancer cells remain single-compartment cells in both models.

How the comparison works live

The browser integrates both models together. Each selected probe produces a **solid spatial curve** and a **dashed point curve** with the same probe color. The comparison checkbox only hides or shows the dashed curves; it does not change the equations.

Core interpretation: similar whole-cell means can coexist with very different local signals. A more detailed spatial model is not automatically more biologically accurate; both require physiological constraints.

Read the reconstructed arbors



Actual interface view in Cell identity mode. Only the three ANN skeletons are drawn. Bridge and balancer cells participate electrically but are selected from menus rather than assigned invented 3D shapes.

Visual / control	Meaning
Branch lines	Traced morphology, not a membrane surface or measured diameter profile.
White ring	Chosen current-injection location.
Colored rings	Local probes; their colors match the associated graph traces.
Gold x	The currently disconnected internal edge, when a cut is active.
Drag / scroll / Show	Rotate, zoom, or filter visible arbors. Hidden cells still participate electrically.
Reset view	Restores viewing angle and zoom only; electrical state is preserved.

In the live identity view, use the **Show** menu to identify a named arbor rather than inferring function from its color. Screen crossings are not connections. The cut and probe picker uses the nearest projected segment within 12 screen pixels and selects its nearer endpoint for a node probe.

Rotate or filter an arbor before selecting a dense crossing. Nearby branches are never joined automatically. The exported trees may omit internal anatomical loops.

Run the model and choose measurements

Control	Effect
Run / Pause	Advance continuously or stop at the current state.
Step 10 ms	Pause continuous running and calculate 20 steps of 0.5 ms each.
Reset to rest	Clear voltages, time, traces and event log. Keep current probes, input site and settings. A checked demo schedules a new pulse.
Speed	Request 10, 50, 200 or 1,000 simulated ms per wall-clock second. The default is 50. Achieved speed can be lower.
Any cell + Add cell mean	Choose any of the 158 cells, including individual bridge and balancer cells.
Group + Add group	Add an ANN, bridge or balancer group mean.
ANN treenode ID + Add node	Probe an original ANN treenode by ID. IDs merged at zero-length edges share one compartment.
Click action: Add local probe	Click a visible ANN branch to record a local compartment.
× beside a probe	Remove that measurement channel. Up to 12 arbitrary probes can be displayed at once.

How averages are calculated

Cell and group means are capacitance weighted: **mean voltage** = $\Sigma(C_i v_i) / \Sigma C_i$. A compartment representing more membrane contributes more to the mean. Equal-capacitance bridge and balancer cells consequently have ordinary arithmetic means within their own groups.

When history starts

Adding or removing probes starts a fresh trace window while preserving voltage and simulation time. New probes have no historical samples. A duplicate probe is not added twice. Changing the visible arbor, color mode or graph window does not change the selected probes.

Switching away from the tab pauses the simulation. The clock is simulated time, not elapsed time on your computer. There is no fixed 160 ms endpoint in LIVE.html.

Choose an input and deliver a pulse

Setting	Default / live limits	Interpretation
Pulse amplitude	1 pA; -20 to +20 pA	Injected current at the chosen compartment. Negative values reverse its direction.
Pulse length	20 ms; 0.5 to 1,000 ms	How long each new pulse lasts; rounded to the fixed 0.5 ms timestep.
Injection site	A reproducible ANN Q1-4 contact node	A synthetic probe location, not an experimentally identified sensory input.

Inject into a particular branch

1. Choose **Move injection site** under Click action.
2. Click a branch; the white ring moves to the selected compartment.
3. Set amplitude and length.
4. Press **Inject now**. If paused, press Run or Step.

Inject into a bridge or balancer cell

Select it under **Any cell**, press **Inject into selected cell**, then Inject now. For an ANN cell, that shortcut chooses its first exported compartment, not an identified soma. Use a branch click for a location you can inspect.

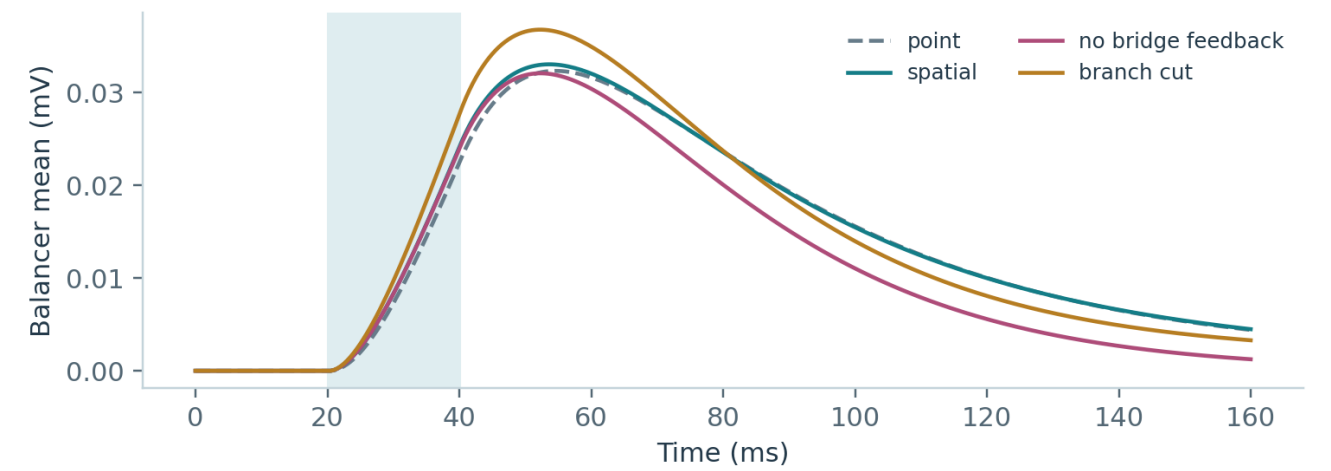
What happens when you change the pulse?

A larger current magnitude generally produces a larger response. A longer pulse supplies current for longer, allowing more temporal integration. Leakage and recurrent interactions determine the eventual trajectory; duration does not imply indefinitely increasing voltage. Near the linear regime, doubling input can approximately double voltage, but saturation and feedback can break that relationship.

Demo timing detail: after reset, the demo starts at 20 ms and uses the current pulse amplitude and length. It ends at 40 ms only with the default 20 ms length. The checkbox's "20-40 ms" label is not dynamically updated when length changes.

A pulse is scheduled at the simulation time when the worker processes the request. Changing the input site does not move an already scheduled pulse. Multiple active pulses add their currents; an extra manual pulse can overlap the demo.

Read the graphs and voltage colors



Baseline reference traces: modeled balancer mean voltage. The shaded region is the original 20-40 ms pulse. This reference figure compares four scenarios; the live graph instead colors curves by your chosen probes.

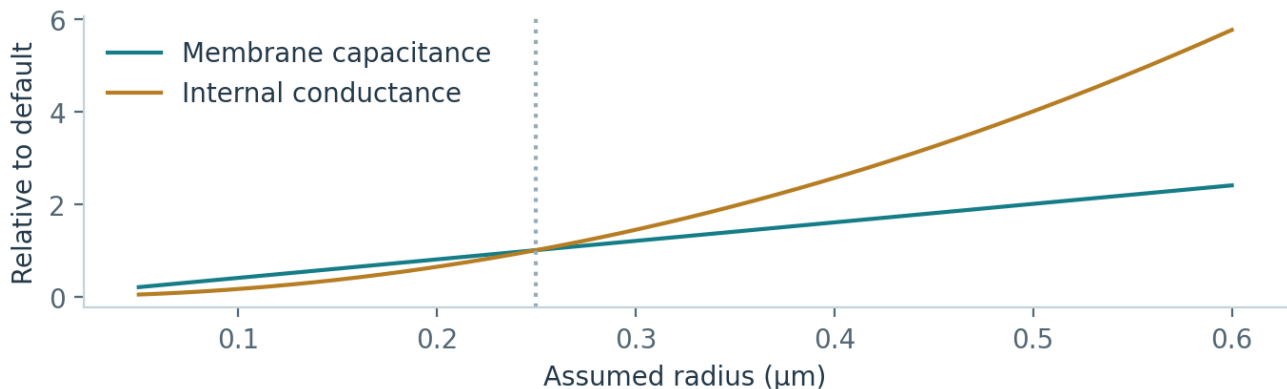
Display	How to read it
Horizontal axis	Simulated time in ms. Window selects 100, 250, 1,000 or 5,000 ms of visible history.
Vertical axis	Voltage deviation from a resting reference, in mV. One automatically scaled axis is shared by all probes.
Solid / dashed	Spatial / equipotential results for the same probe. A point-cell local probe equals its owner's single voltage.
Probe table	Latest numerical spatial and point values; useful when small curves are hidden by a larger response.
Voltage color	Blue: negative. Muted teal: near rest. Gold: positive. The manual $\pm$ range defaults to 1.5 mV; values beyond it clip.
Cell identity color	Anatomical identity rather than activity. Changing it does not change simulation state.

Zero mV means the model's resting reference. It is not an absolute membrane potential of zero. The model has graded electrical responses, not action potentials. A high peak is not a spike count, and balancer voltage is not ciliary beat frequency or swimming speed.

The graph may shift or rescale as new data arrive. Check numeric axes when comparing views. Clearing the trace window discards recorded samples but leaves the electrical state running.

Geometry and internal conduction

Parameter	Default; UI range	Meaning and changes
Cable radius	0.25 μm ; 0.05-2 μm	Uniform assumed radius. Increasing it increases membrane area and capacitance linearly, but internal conductance quadratically. Decreasing it does the reverse.
Axial resistivity	150 $\Omega\cdot\text{cm}$; 1-2,000 $\Omega\cdot\text{cm}$	Resistance of the internal material. Higher values weaken internal coupling; lower values strengthen it. Local gradients can become more prominent with weaker coupling.
Internal conductance $\times$	1; 0-10	Multiplier on ANN internal conductances only. Higher values couple compartments more strongly without adding membrane. Zero removes internal cable coupling.



Analytic scaling used by the code, relative to the default radius. Specific capacitance and resistivity are held fixed in this illustration.

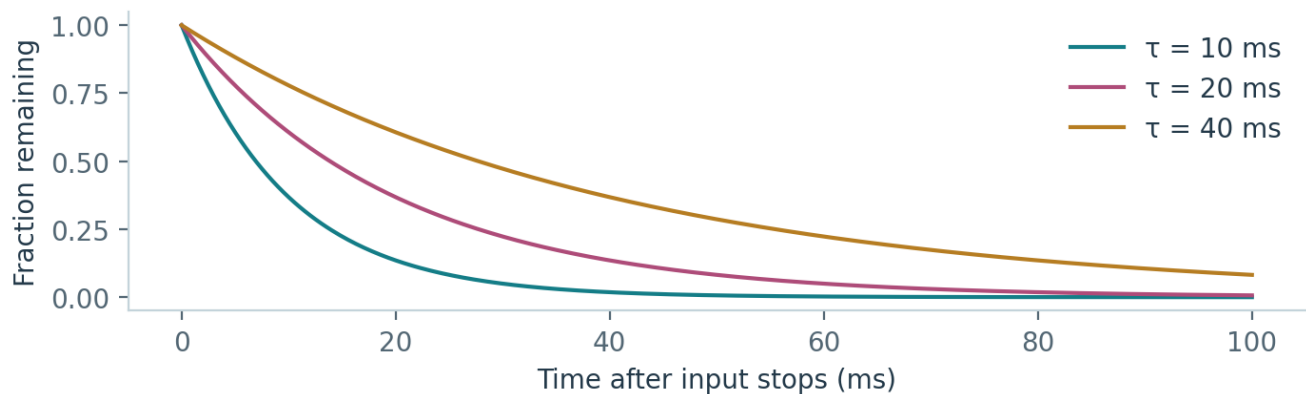
Why radius has no simple “more means higher voltage” rule

Doubling radius doubles capacitance and total leak at fixed leak time constant, but quadruples internal conductance. Current spreads more effectively while charging more membrane. A fixed input may therefore yield a smaller local voltage even as coupling becomes stronger. Network feedback can alter the final outcome.

The UI ranges are allowed exploration bounds, not experimentally supported confidence intervals. Strong internal coupling approaches the equipotential limit. Zero coupling is an extreme mathematical intervention, not an intact neuron.

Membrane charge storage and leakage

Parameter	Default; UI range	Meaning and changes
Specific capacitance	1 $\mu\text{F}/\text{cm}^2$; 0.1-5	Capacitance per ANN membrane area. Increasing it increases charge storage and, at fixed tau, total leak. Fixed-current voltage responses generally shrink.
Leak time constant	20 ms; 1-200 ms	Passive relaxation time τ . Larger τ means weaker leak at fixed C, usually longer persistence and larger sustained-current responses. Smaller τ gives stronger leak.
Bridge / balancer C	0.5 pF; 0.01-10 pF	Capacitance of each bridge and balancer cell in both models. Increasing it generally reduces their response to a fixed current. ANN point capacitance is derived separately from its arbor.



An isolated passive compartment after input stops: $v(t) / v(0) = \exp(-t / \tau)$. Incoming currents make a connected circuit's decay more complicated.

The coupling between capacitance and leak

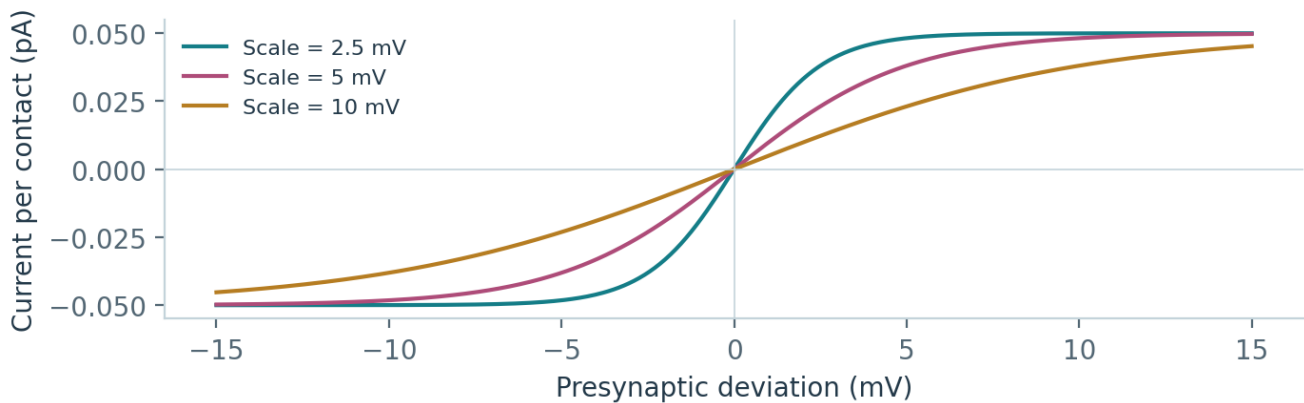
The implementation sets leak conductance to C / τ . If you increase capacitance while holding τ fixed, you also increase leak conductance. You are not changing capacitance alone. For an isolated compartment with constant input, the steady deviation is $v = I\tau / C$.

During a live parameter update, voltage is retained. If capacitance changes, stored charge $C \times v$ changes too. This is an artificial intervention; use Reset for comparisons that should start from a common resting state.

Synaptic strength, sensitivity and sign

Parameter	Default; UI range	Meaning and changes
Current scale / contact	0.05 pA; 0-1 pA	Saturating current magnitude per anatomical contact. Larger values strengthen interactions. Zero disables all modeled synaptic currents.
Synaptic voltage scale	5 mV; 0.1-50 mV	Divisor inside tanh. A larger scale weakens sensitivity near rest; a smaller scale makes small deviations exert stronger effects and reach saturation sooner.
All-contact sign	+1; +1 or -1	Common sign for every contact. -1 reverses the presynaptic voltage-to-current relationship; it does not assign measured neurotransmitter identities.

Per-contact current = sign × current scale × $\tanh(v_{\text{pre}} / \text{voltage scale})$. Contributions from each measured contact add at the receiving compartment.



The same positive current scale with three voltage scales. Each curve is a hypothetical contact-response function, not measured physiology.

At $v_{\text{pre}} = 0.5$ mV with default settings, one contact contributes about **0.005 pA**, not a constant 0.05 pA. The tanh function saturates smoothly between -1 and +1. Ten contacts at the same presynaptic voltage contribute ten times one contact because equal strength is assumed.

Positive sign is labeled positive coupling: positive deviations produce positive current, but negative deviations produce negative current. This signed approximation is neither a nonnegative firing rate nor a receptor conductance model. Negative sign does not guarantee every network response becomes smaller.

Feedback, cuts and live interventions

Intervention	Exactly what changes	What remains
Disable bridge → ANN feedback	44 returning contacts are disabled; 236 of 280 contacts remain active. Press Apply to activate the checkbox change.	Bridge cells, ANN → bridge contacts, other contacts and membrane compartments.
Click to select a cable cut	The selected internal edge is immediately disconnected. Choosing another cut restores the previous one.	Capacitance and synaptic contacts on both sides; indirect network paths may remain.
Use balanced default cut	Selects the preset central-arbor edge. Press Apply to activate it. The represented cable-length split is about 55% / 45%.	No tissue is removed, and the two portions are not modeled as dying or healing.
Uncheck Cut + Apply	Restores the selected internal edge.	The current voltage state is preserved; any earlier perturbation effects can persist.

Interpret a larger signal after a cut

A local voltage can rise because injected current has fewer internal paths through which to spread. Downstream means can rise or fall depending on contact locations and recurrent activity. A larger signal does not mean improved biological function.

Apply, reset, and restore defaults are different

Apply parameters now changes subsequent dynamics while retaining voltages and history, and logs the intervention. **Reset to rest** starts from zero deviation using the current controls. **Restore defaults & reset** restores default electrical and pulse settings, the default input site, feedback and no cut; current probes remain.

The point model shares membrane settings, contact changes and current inputs but cannot represent an internal local cable cut. Exported skeleton trees may omit alternate anatomical routes, which makes cut predictions especially uncertain.

The equations behind each step

For compartment i , voltage v is measured relative to the resting reference. The model balances membrane storage, leak, internal cable currents, directed contact currents and the external input:

$$C_i \, dv_i/dt = -(C_i/\tau)v_i + \sum_j g_{ij}(v_j - v_i) + \sum_{k \rightarrow i} s_{I_{syn}} \tanh(v_k / V_{scale}) + I_{ext,i}(t)$$

Term	Interpretation
$C_i \, dv_i/dt$	Current required to change voltage in a compartment with capacitance C_i .
$-(C_i / \tau)v_i$	Leak pulls the deviation toward zero.
$g_{ij}(v_j - v_i)$	Internal current flows according to the voltage difference between neighboring pieces of one ANN.
Synaptic sum	One directed contribution per measured contact. Autapses and contacts sharing a presynaptic connector are retained.
I_{ext}	A chosen pulse; it is not a measured sensory drive.

From skeleton geometry to electrical coefficients

Each node receives half the length of every incident edge, denoted l_i . For radius r , specific capacitance C_m , axial resistivity R_a and edge length L_{ij} :

$$C_i = 0.01 \times C_m \times 2\pi r l_i \text{ [pF]}$$

$$g_{ij} = \text{axial_scale} \times \pi r^2 \times 10^5 / (R_a L_{ij}) \text{ [nS]}$$

Lengths are μm ; C_m is $\mu\text{F}/\text{cm}^2$; R_a is $\Omega \cdot \text{cm}$. The cylinder approximation uses lateral surface area, without separately reconstructed soma surfaces or end caps. Three zero-length parent-child edges are merged before solving.

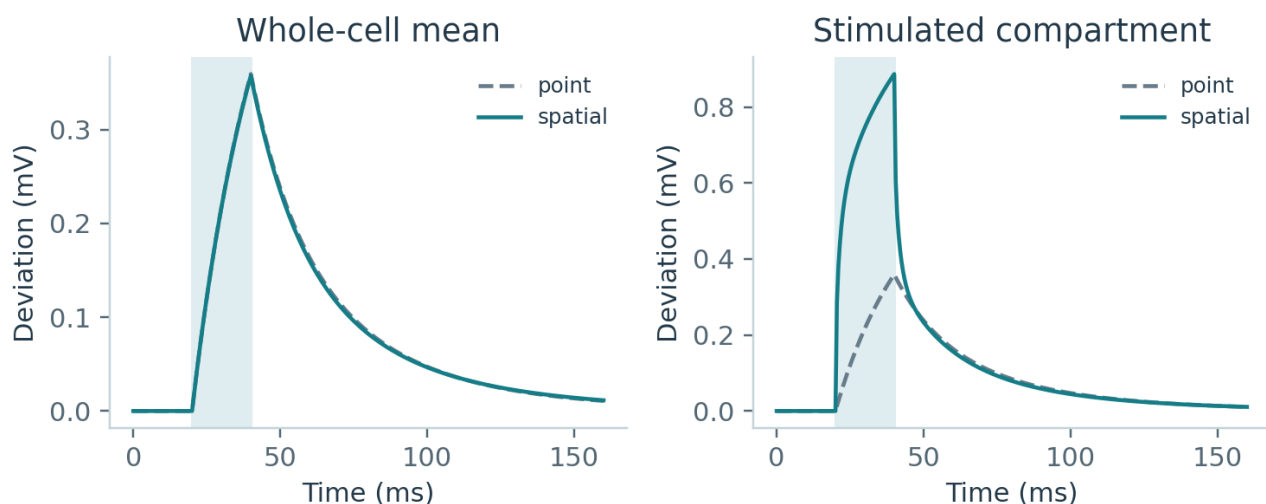
Numerical method

Passive leak and internal currents use backward Euler; synaptic currents use the previous step's voltage. A direct forest solve exploits the tree-shaped internal skeletons. The synaptic network itself can be recurrent. Both spatial and point models integrate at a fixed 0.5 ms timestep in the live app.

What the default results show

These are the saved original baseline runs: 1 pA at the default central ANN site from 20 to 40 ms, radius 0.25 μm and positive contact sign. Values are peak absolute deviations in mV; different readouts can peak at different times.

Readout	Point	Spatial	No feedback	Cable cut
ANN Q1-4 mean	0.35929	0.35763	0.35312	0.35359
ANN Q1Q2 mean	0.01018	0.00984	0.00189	0.01561
ANN Q3Q4 mean	0.02336	0.02083	0.01852	0.01975
Bridge mean	0.12360	0.11329	0.10426	0.04162
Balancer mean	0.03234	0.03305	0.03208	0.03679
Stimulated location	0.35929	0.88656	0.88356	1.19761

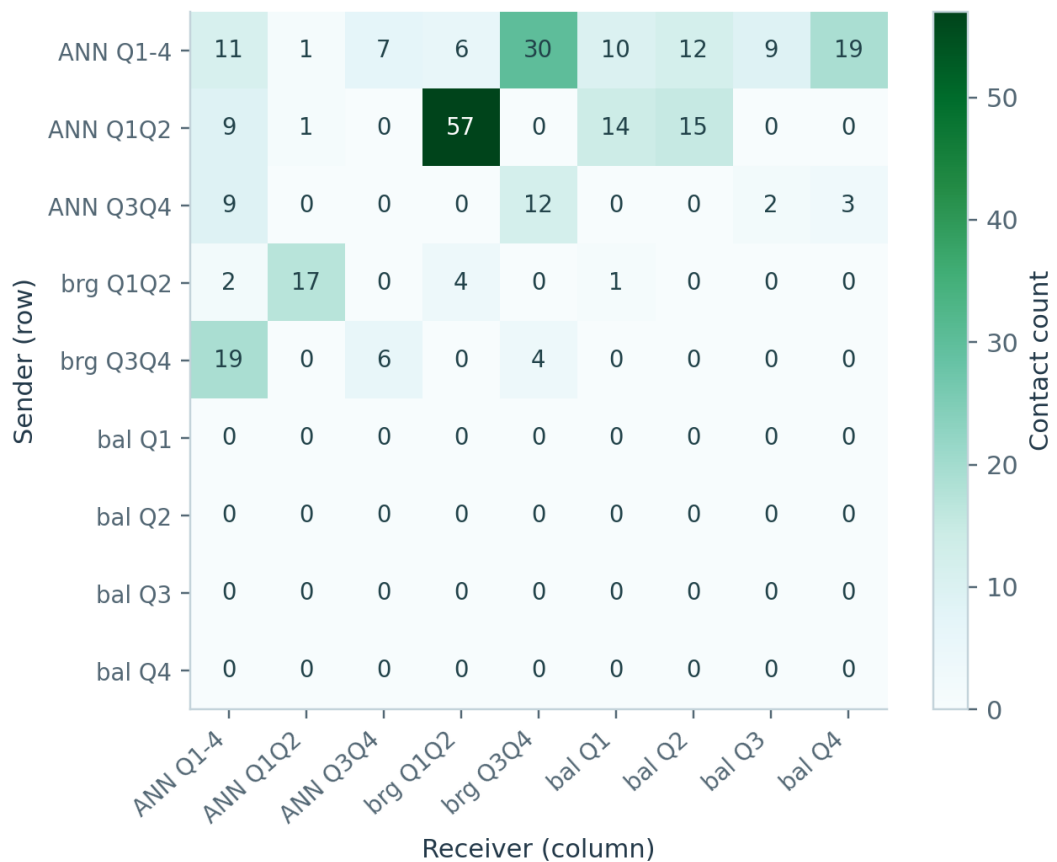


The central cell means are very similar, but the stimulated compartment differs substantially. The two panels deliberately have different voltage-axis ranges; compare their numbers, not just curve height.

Feedback removal strongly reduces the ANN Q1Q2 mean peak in this preset. The default cut reduces the bridge mean peak but increases the local and balancer peaks. Those outcomes follow from these equations, inputs and assumed parameters; they do not establish a biological function of injury or feedback.

The useful research question is whether a local measurement can distinguish plausible mechanisms when a cell average cannot. The current results motivate that question; they do not answer it experimentally.

Read the measured wiring matrix



Grouped source matrix from Figure 3 data. **Rows send; columns receive.** Every entry counts directed anatomical contacts, and all entries sum to 280. ANN labels are individual neurons; brg and bal labels aggregate cells by anatomical quadrant.

Example	Meaning
ANN Q1Q2 → brg Q1Q2 = 57	57 contacts from that ANN to the bridge group, not 57 neurons or spikes.
brg Q1Q2 → ANN Q1Q2 = 17	A returning anatomical connection, with a different number of contacts.
A zero entry	No contact included for that group-to-group direction in the selected table; not proof of no biological influence.
A diagonal entry	Within a multi-cell group it can connect different cells; within one ANN it is a self-directed contact (autapse).

The original matrix is unchanged by simulation controls. Disabling feedback changes which contacts contribute current, not the measured anatomical table. Contact count becomes proportional to functional strength only through the model's equal-strength assumption.

Sensitivity analysis and summary metrics

The original **results/sensitivity.csv** contains 18 scenarios: three radii (0.1, 0.25, 0.5 μm), two uniform signs (-1, +1), and three current scales (0, 0.05, 0.1 pA). Zero-current scenarios repeat for both signs as controls. This is scenario analysis, not biological replicates or confidence bounds.

Metric	Meaning
mean_trace_rmse_mV	Root mean squared difference between point and spatial traces over time and the five mean readouts. The stimulated local readout is excluded. It measures model disagreement, not error against experimental truth.
peak_abs_mV	Largest absolute deviation during the recorded run. Magnitude alone does not tell you whether the peak was positive or negative.
peak_time_ms	When that absolute peak occurs. It is not a direct measurement of conduction velocity.
integral_mV_ms	Signed area under a trace: combines amplitude and duration. Positive and negative portions can cancel.

A slice through the saved sensitivity results

Radius (μm)	Sign / scale	Mean-trace RMSE (mV)
0.10	+1 / 0.05 pA	0.040291
0.25	+1 / 0.05 pA	0.003460
0.50	+1 / 0.05 pA	0.000881

This particular slice shows smaller model disagreement at larger radii. It does not establish a universal monotonic rule across inputs, signs, gains or unknown geometry.

Why mean traces agree when synapses are off

Internal currents redistribute charge without creating it. With total capacitance and uniform leak time constant matched, the capacitance-weighted passive cell mean follows the same equation in both representations. Local spatial voltages can still differ. The near-zero mean-trace discrepancy in this control is a useful implementation check.

Experiment 1: local signal versus cell mean

Question

Can two locations on one ANN respond differently while the whole-cell mean looks nearly unchanged?

Set up the measurements

1. Restore defaults and reset. Keep the default input site and use the default 1 pA, 20 ms pulse.
2. Add ANN Q1-4 cell mean.
3. Add a local probe at or near the white input ring, and a second on a distant part of the same arbor. Use Show and rotation to avoid ambiguous crossings.
4. Keep the equipotential comparison visible.

Run and record

5. Reset after selecting all probes, so all channels record from the beginning.
6. Run beyond the demo pulse, then pause at a chosen common endpoint, for example 160 ms.
7. Export CSV and JSON. Record the actual treenode IDs and endpoint; spatial distance on the screen alone is not a reproducible probe definition.

What to compare

Compare local peak magnitudes, timing and decay against the mean. The dashed point curves for both local locations must be identical because they belong to one point cell. The solid local curves may differ. In the original preset, the input-site peak differs more between representations than the central mean does.

Keep fixed	Record
Input location, amplitude and duration	The input treenode ID and pulse event.
All parameters and feedback/cut state	Exported parameter values and topology events.
Probe set and time window	Exact IDs, model labels, timestep and endpoint.

A small mean difference does not prove spatial effects are absent. Conversely, a local difference in an assumed model is not evidence that the real cell exhibits that difference.

Experiments 2 and 3: coupling and perturbations

Experiment 2: vary internal resistivity

Use the same input and probes as Experiment 1. Run separate reset trials at, for example, 75, 150 and 300 $\Omega\cdot\text{cm}$. Change only axial resistivity, press Apply, reset, and use the same recording endpoint. Export each trial with a descriptive filename.

Lower resistivity strengthens internal coupling; higher resistivity weakens it. Compare the difference between the two local traces and their agreement with the point model. A stronger local gradient is plausible with weaker coupling, but the full network response is not guaranteed to change monotonically.

Experiment 3: returning contacts versus a local cut

- A. Intact control:** defaults, feedback on, cut off; reset and record.
- B. Feedback removal:** feedback off, cut off; Apply, reset, and repeat.
- C. Local cut:** feedback on; choose a specific edge, then reset and repeat.
- D. Recovery control:** uncheck Cut and Apply; reset and repeat the intact trial.

For these trials, add one or more individual bridge/balancer cells or group means before recording. Adding them during an ongoing trial clears the trace window. The balanced preset is reproducible, but an arbitrary cut should be identified by its logged edge index and the bundled anatomy version.

Separate state history from parameter effects

Changing a parameter in mid-run is useful for an intervention experiment. It is a different question from comparing two models started at rest. Live updates retain voltage, and previous pulses or cuts can still affect the next response. Use resets for clean matched-start comparisons.

Interpretation: feedback removal tests a specified contact subset. A cut tests an internal route in the exported skeleton. Neither intervention establishes cell death, physiological inhibition, behavioral improvement or biological causality.

Before making a research claim

Repeat across plausible parameter settings and multiple probe/input locations. Check timestep convergence for the settings used. A defensible next study would calibrate competing models against the same observations, then test a held-out perturbation that separates them.

Export, retain and reproduce your work

The browser retains at most **10,000 recent integration samples**, about 5,000 ms at 0.5 ms per step. This is a rolling buffer, not an unlimited recorder. The visible graph window and the retained buffer are separate settings.

File / action	Contents and purpose
Export traces CSV	Time plus every selected probe's spatial and point values in the retained window.
Export experiment JSON	Initial/current parameters, interventions, probe-set history, source identifier and retained samples. Not a resumable full-state checkpoint; no session import is provided.
LIVE.html	Self-contained live simulator.
EXPLORE.html	Original fixed four-scenario replay and grouped wiring matrix.
live/solver.js; worker.js; app.js	Numerical kernel, worker messaging and interface.
build_live.py; live/anatomy.json	Rebuilds the live page from bundled source anatomy; derived model data.
combcircuit.py; results/	Python reference, original traces, contact table, audit, metrics and sensitivity analysis.
data/provenance.json; data/upstream/	Exact source files, URLs and hashes.
docs/; tests/	Model, audit, study plan, transcript and verification code.

Optional rebuild from the project folder

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python build_live.py
python -m unittest discover -s tests -v
```

Python 3.11 or newer is recommended by the project. Package installation requires internet; subsequent calculations use bundled data. The JavaScript/Python comparison test additionally needs Node.js and is explicitly skipped without it. Node is not needed to use LIVE.html. On Windows, activate with `.venv\Scripts\activate`.

Do not confuse rebuild commands: `python combcircuit.py` regenerates the original fixed comparison; `python build_live.py` regenerates LIVE.html. Save exports before Reset, Clear trace window, or a probe-set change.

Live versus legacy settings

Python parameter	Default	Live 0.2 equivalent
radius_um	0.25	Cable radius
axial_resistivity_ohm_cm	150	Axial resistivity
capacitance_uF_cm2	1	Specific capacitance
leak_tau_ms	20	Leak time constant
point_capacitance_pF	0.5	Bridge / balancer C; not the ANN point-cell capacitance
synaptic_current_pA	0.05	Current scale / contact
synaptic_scale_mV	5	Synaptic voltage scale
synaptic_sign	+1	All-contact sign
axial_scale	1	Internal conductance $\times$
pulse_pA	1	Pulse amplitude
pulse_start_ms	20	Demo starts at 20 ms; manual pulses start at the current simulation time
pulse_end_ms	40	Demo end is 20 ms + selected length; manual pulse end is start + length
duration_ms	160	No fixed endpoint in LIVE.html; pause or reset to stop
dt_ms	0.5	Fixed in the live interface; alter the code for numerical convergence work

What remains fixed in EXPLORE.html

The legacy page replays four precomputed cases and six readouts. Its slider uses 5 ms morphology snapshots, while its traces use 0.5 ms steps. Selecting a scenario emphasizes one curve; it does not rerun the model. Download exports all six readouts of that selected case.

The legacy viewer contains hardcoded timing assumptions and a nonnegative voltage color map. If Python timing or negative-response settings are changed, use the CSV output or update that viewer code. LIVE.html has signed colors and dynamic time axes, but retains the fixed 0.5 ms integration step.

Troubleshooting and common misreadings

What you notice	Explanation / action
The graph is flat at the start	Before the demo begins at 20 ms, the model is at rest. Run or Step through the pulse, or inject manually.
Inject now did nothing while paused	The pulse is queued at the current simulated time. Advance with Run or Step to see its effect.
Clicking selected the wrong branch	Overlapping projections are ambiguous. Rotate, zoom, or Show one arbor, then click again.
A small response seems invisible	All traces share one y axis. Check the table, remove a much larger probe, or export and plot separately.
Colors hardly change	The voltage range may be large relative to the response. Adjust $\pm$ range; this changes display only.
Changing a number had no effect	Model parameter fields require Apply or Reset. Pulse fields affect newly queued pulses. Selecting a cut by branch click applies it immediately.
History disappeared	Adding/removing probes starts a new trace window. Reset and Clear also discard retained samples. Export first.
The run pauses when switching tabs	This is intentional. Return to the tab and press Run.
The requested speed is not reached	Device speed, rendering and probe count affect pacing. The app reports achieved speed and does not skip integration steps.
LIVE.html remains at Preparing	Use a browser with JavaScript and Web Workers enabled. Re-extract the complete ZIP; Chromium was tested. If errors persist, capture the browser console message.

Avoid these conclusions

A bright branch is not a spike. A larger voltage is not automatically better function. A zero matrix entry is not proof of no influence. A point/spatial disagreement is not error against experimental truth. Passing tests is not biological validation.

If strong gains or unusual settings produce implausible behavior, reset and return to a known baseline. The allowed UI range is not a guarantee of physiological realism or numerical convergence for every setting.

Glossary and units

Term	Meaning
Arbor	The branching structure traced for a neuron.
ANN	Aboral nerve net; the three modeled ANN neurons have explicit morphology.
Syncytial	A continuous cellular structure containing multiple nuclei. This does not establish which internal loops are present in the exported skeleton.
Compartment	A computational piece of membrane and its local voltage.
Equipotential	Represented by a single voltage throughout a cell.
Capacitance	Charge storage per voltage change; $Q = C \times v$.
Conductance / resistivity	Ease of current flow / resistance property of a material.
Leak time constant	Passive relaxation timescale; here $\tau = C / \text{leak conductance}$.
Autapse	A synaptic contact whose sender and receiver belong to the same cell.
Feedback	A returning influence; here the toggle specifically removes bridge → ANN contacts.
Treenode / connector	A morphology skeleton point / a contact identifier linking annotated endpoints.
RMSE	Root mean squared difference. Its interpretation depends on the reference being compared.

Unit	Quantity	Unit	Quantity
μm	Length: micrometer	ms	Time: millisecond
mV	Voltage: millivolt	pA	Current: picoampere
pF	Capacitance: picofarad	nS	Conductance: nanosiemens
$\Omega\cdot\text{cm}$	Axial resistivity	$\mu\text{F}/\text{cm}^2$	Capacitance per membrane area

In these units, $\text{pF} \times \text{mV}/\text{ms} = \text{pA}$, and $\text{nS} \times \text{mV} = \text{pA}$. These relationships let the model balance all current terms consistently.

What is verified, and what remains unknown?

Implementation checks

The live release reports **10 passing numerical test methods**. Its new full-state comparison covers nine scenarios against a SciPy sparse solve: default, feedback off, balanced cut, arbitrary cut, negative input, thin cable with negative contact sign, zero input, individual bridge input, and a parameter change at 40 ms. Maximum observed state discrepancy was about **1.46×10^{-10} mV**.

Other tests cover source hashes, direction/counts, zero-length merging, matched capacitance, internal charge conservation, passive mean behavior, strong-coupling limits, default timestep refinement and perturbation implementation. Browser checks covered live continuation, parameter updates, individual-cell input, branch picking, cuts, exports, reset and mobile layout.

Data audit

Pinned source tables contain 1,132 endpoint records and 430 connector IDs. Fifty-two postsynaptic records lack presynaptic partners and are excluded rather than assigned guessed partners; one connector has only a presynaptic endpoint. The selected circuit retains 280 contacts, including an autapse and a bridge-to-balancer contact.

The GitHub tables are pinned to a specific commit; CATMAID arbors were retrieved separately on 2026-09-27. Modeled ANN contact-node IDs were checked, but this does not prove the two snapshots are identical versions. Source radii are largely unspecified. Three zero-length edges were merged, and no missing paths or synapses were invented.

Scientific boundaries

The model lacks active ion channels, action potentials, calcium reporting, ciliary mechanics, behavior, learning and physiological fitting. It uses a selected circuit, uniform electrical assumptions, and tree exports that may miss syncytial loops. Longer live runs do not automatically validate long-term dynamics. Default convergence checks do not certify every extreme setting.

Next research step: constrain geometry and membrane/contact properties, fit competing models to the same observations, then evaluate a held-out perturbation. A useful prediction should survive uncertainty and be distinguishable by a feasible measurement.

Sources, credits and companion video

[1] Original anatomical study. Jokura and colleagues, Neural connectome of the ctenophore statocyst, eLife. elifesciences.org/articles/108420.

[2] Authors' source data and code. JekelyLab/Jokura_2024_ctenophore_apical_organ. [GitHub repository](#). The release pins commit ccdc5b2d8e897dcb4d1df1d4a651b6925844320a and separately records the CATMAID snapshot in data/provenance.json.

[3] This guide's implementation source. CombCircuit Live 0.2: live/solver.js, live/app.js, build_live.py, combcircuit.py, bundled documentation and result files. The PDF explains the delivered code; it does not report a new biological experiment.

Attribution. The project did not map these cells. Original anatomical credit belongs to the researchers. See LICENSE and THIRD_PARTY_NOTICES.md for code and upstream terms. Original researchers have not endorsed this workbench.

Video chapter shortcuts

Time	Topic	Time	Topic
00:00	Introduction / opening the app	06:38	Radius
00:41	Biological circuit	07:22	Internal resistance / coupling
01:20	Arbors and compartments	08:05	Capacitance / leak
02:07	Spatial versus point models	08:48	Synaptic response
02:51	Run, pause, step, reset	09:33	Signs / feedback
03:40	Local probes	10:18	Cable cuts
04:22	Any cell / group / treenode	11:05	Controlled experiment
05:02	Move input / inject	11:47	Exports
05:50	Graphs and colors	12:35	Legacy matrix / figures
		13:21	Numerics and scientific limits

Companion: **CombCircuit-Live-Walkthrough.mp4**, about 14 minutes, with synthetic narration, burned-in captions and 19 embedded chapters. A transcript, subtitle file and chapter guide are included in the project.

A productive question to keep asking: which local measurement or perturbation would distinguish two plausible models, after both have been constrained by the same evidence?